

Introducing Markov Decision Process

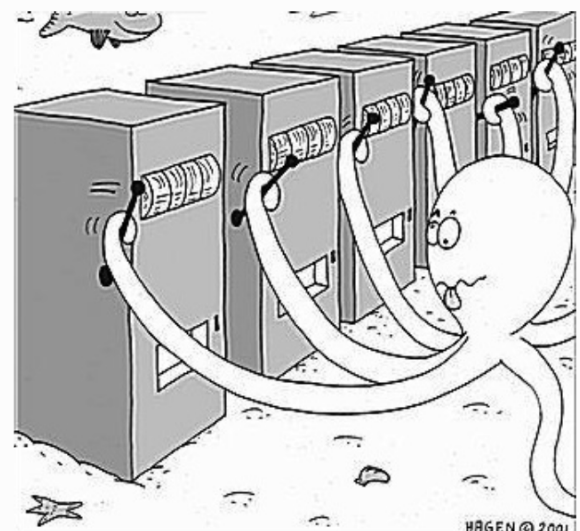
Markov decision process (MDP) is the central modeling object in RL, which represents a model of interactive decision-making.

Markov decision process
↑
decision-making
+
Markov property
+
stochastic process

Markov decision process
↑
contextual bandits
↑
multi-armed bandits

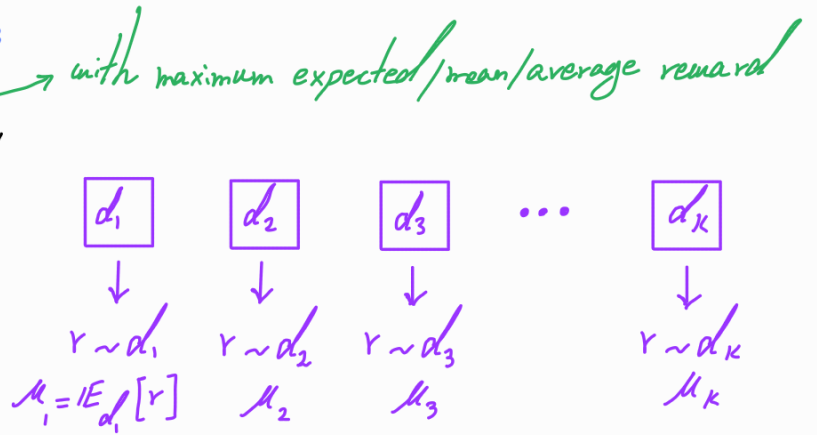
Bandits to MDP

Bandits represent interactive decision-making settings of varying complexity. The core challenge of exploration versus exploitation is present in all these settings. Bandit algorithms are designed to enable strategic exploration.



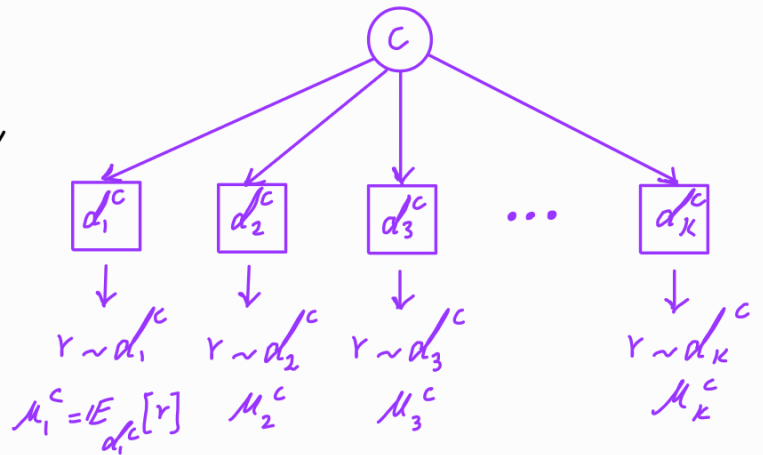
Multi-armed bandit (MAB):

Learning an optimal action/
decision/arm through
interaction



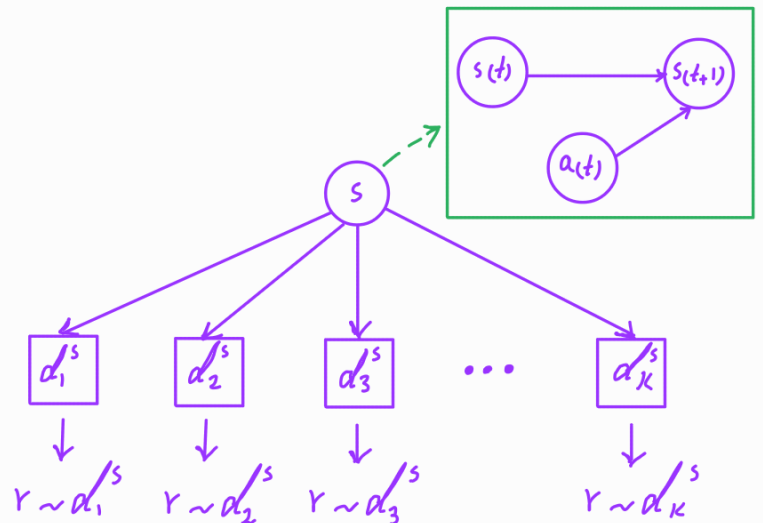
Contextual bandit:

Learning an optimal action/
decision/arm per context
through interaction



Markov decision process (MDP):

Learning an optimal action/
decision per evolving state
through interaction,
where the state evolution
depends on action/decision



Markov Models

→ the setting without reward;
reward can be added to all of them

- Markov chain (MC)

$$(\mathcal{S}, P)$$



$\mathcal{S}$: state space, set of all possible states

P : transition function (matrix); $P: \mathcal{S} \rightarrow \Delta(\mathcal{S})$

$$\begin{aligned} \mathcal{S} &= \{0, 1\} \\ \Delta(\mathcal{S}) &= \begin{bmatrix} \times & \times \\ \times & \times \end{bmatrix} \\ s=0 & \quad s=1 \\ & \quad \quad \quad \rightarrow [0.3 \ 0.7] \end{aligned}$$

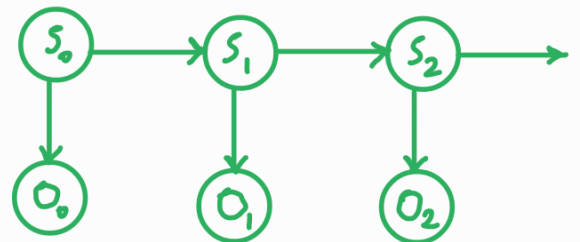
$$P(s'|s) = P[s_{t+1}=s' | s_t=s] \rightarrow \text{probability of transitioning from } s \text{ to } s'$$



$$\begin{matrix} \mathcal{S} \times \mathcal{S} \\ \downarrow \\ s_0 \xrightarrow{P(s_1|s_0)} s_1 \xrightarrow{P(s_2|s_1)} s_2 \dots \\ \downarrow \quad \downarrow \\ \in \mathcal{S} \quad \in \mathcal{S} \end{matrix}$$

- Hidden Markov Model (HMM)

$$(\mathcal{S}, P, \mathcal{O}, \mathcal{Z})$$



$\mathcal{S}$: state space

P : transition function (matrix); $P: \mathcal{S} \rightarrow \Delta(\mathcal{S})$

$\mathcal{O}$: observation space, set of all possible observations

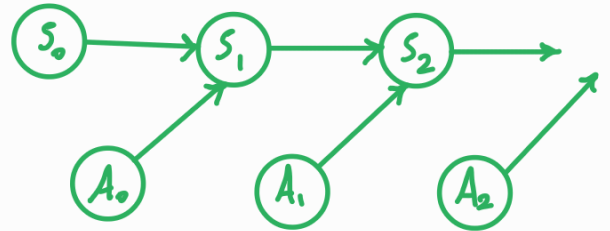
$\mathcal{Z}$: observation function (emission probabilities); $\mathcal{Z}: \mathcal{S} \rightarrow \Delta(\mathcal{O})$

$$\mathcal{Z}(o|s) = P[o_t=o | s_t=s] \rightarrow \text{probability of observing } o \text{ at } s$$

$$\begin{matrix} s_0 \xrightarrow{\mathcal{Z}(o_0|s_0)} o_0 \xrightarrow{P(s_1|s_0)} s_1 \xrightarrow{\mathcal{Z}(o_1|s_1)} o_1 \xrightarrow{P(s_2|s_1)} s_2 \dots \end{matrix}$$

- Markov Decision Process (MDP)

$$(\mathcal{S}, \mathcal{A}, P)$$

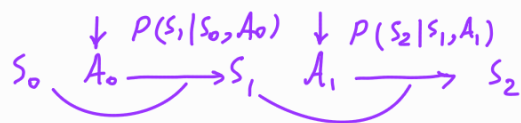


$\mathcal{S}$: state space

$\mathcal{A}$: action space, set of all possible actions

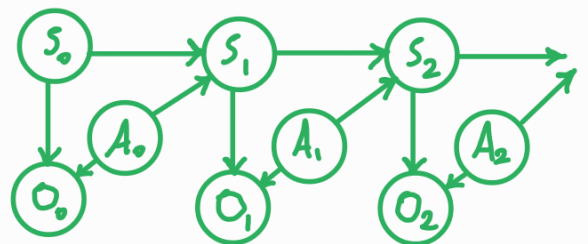
P : transition function ; $P : \mathcal{S} \times \mathcal{A} \longrightarrow \Delta(\mathcal{S})$

$$P(s' | s, a) = P[s_{t+1} = s' | s_t = s, A_t = a]$$



- Partially Observable Markov Decision Process (POMDP)

$$(\mathcal{S}, \mathcal{A}, P, \mathcal{O}, \mathcal{Z})$$



$\mathcal{S}$: state space

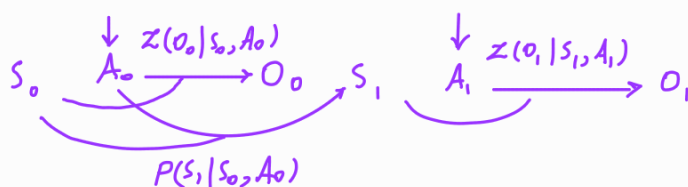
$\mathcal{A}$: action space

P : transition function ; $P : \mathcal{S} \times \mathcal{A} \longrightarrow \Delta(\mathcal{S})$

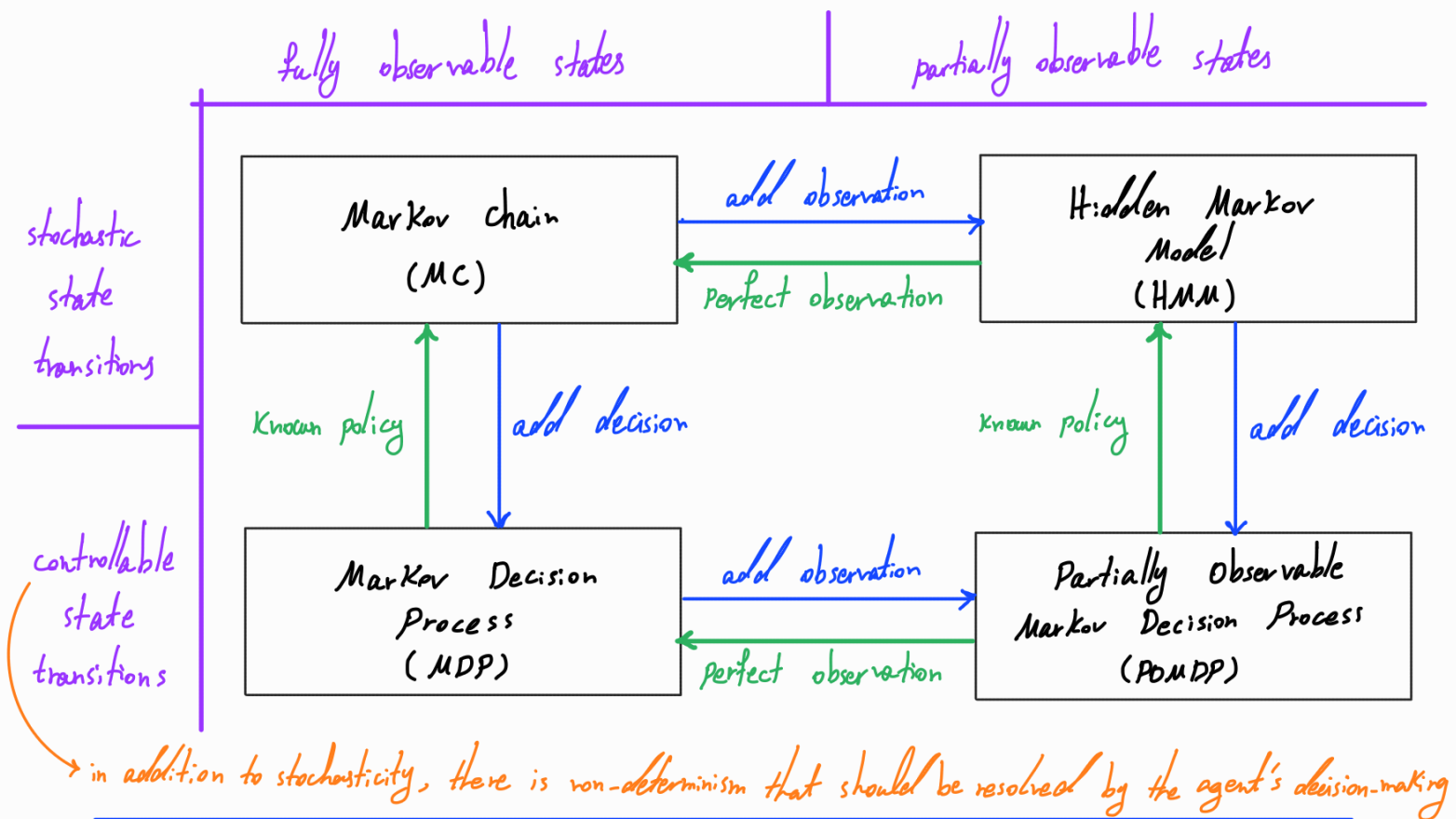
$\mathcal{O}$: observation space

$\mathcal{Z}$: observation probabilities ; $\mathcal{Z} : \mathcal{S} \times \mathcal{A} \longrightarrow \Delta(\mathcal{O})$

$\mathcal{Z}(o | s, a) \sim$ probability of observing o at s taking a



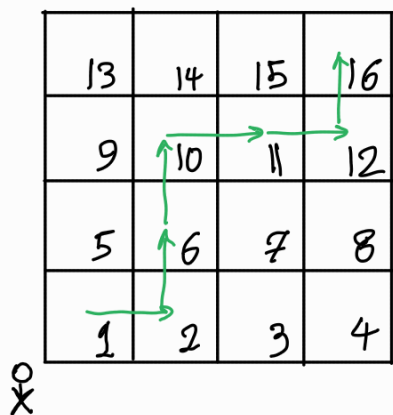
Overview of Markov Models



Example:

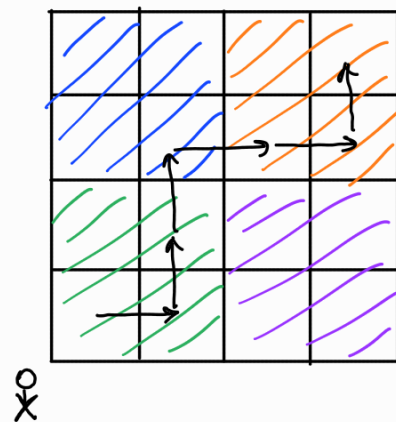
MC

$\begin{cases} \text{goes right} & 0.6 \\ \text{goes up} & 0.4 \end{cases}$
 $S = \{1, 2, \dots, 16\}$



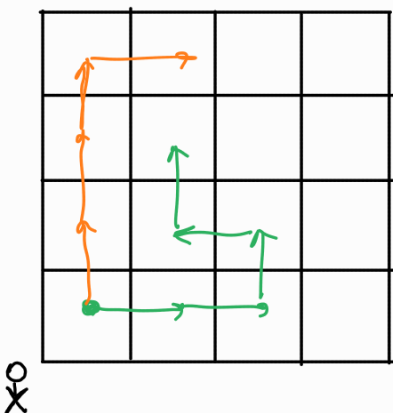
HMM

$\begin{cases} \text{goes right} & 0.6 \\ \text{goes up} & 0.4 \end{cases}$
 $S = \{1, 2, \dots, 16\}$
 $O = \{\bullet, \circ, \bullet, \circ\}$



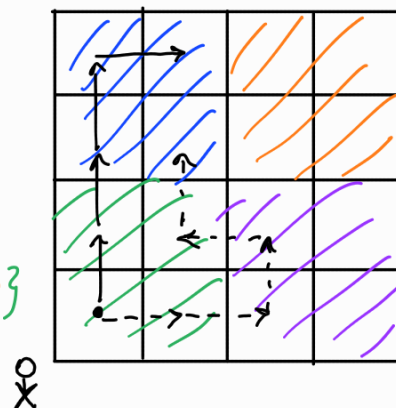
MDP

$S = \{1, 2, \dots, 16\}$
 $A = \{r, l, u, d, s\}$



POMDP

$S = \{1, 2, \dots, 16\}$
 $A = \{r, l, u, d, s\}$
 $O = \{\bullet, \circ, \bullet, \circ\}$



Example: Entertainment system

states of human user $S = \{\text{like}, \text{dislike}\}$

system's observations of human $O = \{\text{smile}, \text{frown}, \dots\}$

system's actions $A = \{\text{content 1}, \text{content 2}, \dots, \text{content } k\}$